| Lossless | Lossy |
|---|---|
| run-length | CS&Q |
| Huffman | JPEG |
| delta | MPEG |
| LZW | |

a. Lossless or Lossy

| Method | Group size: input | output |
|---|---|---|
| CS&Q | fixed | fixed |
| Huffman | fixed | variable |
| Arithmetic | variable | variable |
| run-length, LZW | variable | fixed |

b. Fixed or variable group size

TABLE 27-1
Compression classifications. Data compression methods can be divided in two ways. In (a), the techniques are classified as *lossless* or *lossy*. Lossless methods restore the compressed data to exactly the same form as the original, while lossy methods only generate an approximation. In (b), the methods are classified according to a *fixed* or *variable* size of group taken from the original file and written to the compressed file.

Images transmitted over the world wide web are an excellent example of why data compression is important. Suppose we need to download a digitized color photograph over a computer's 33.6 kbps modem. If the image is not compressed (a *TIFF* file, for example), it will contain about 600 kbytes of data. If it has been compressed using a *lossless* technique (such as used in the *GIF* format), it will be about one-half this size, or 300 kbytes. If *lossy* compression has been used (a JPEG file), it will be about 50 kbytes. The point is, the download times for these three equivalent files are 142 seconds, 71 seconds, and 12 seconds, respectively. That's a big difference! JPEG is the best choice for digitized photographs, while GIF is used with *drawn* images, such as company logos that have large areas of a single color.

Our second way of classifying data compression methods is shown in Table 27-1b. Most data compression programs operate by taking a group of data from the original file, compressing it in some way, and then writing the compressed group to the output file. For instance, one of the techniques in this table is **CS&Q**, short for **coarser sampling and/or quantization**. Suppose we are compressing a digitized waveform, such as an audio signal that has been digitized to 12 bits. We might read two adjacent samples from the original file (24 bits), discard one of the sample completely, discard the least significant 4 bits from the other sample, and then write the remaining 8 bits to the output file. With 24 bits in and 8 bits out, we have implemented a 3:1 compression ratio using a lossy algorithm. While this is rather crude in itself, it is very effective when used with a technique called *transform compression*. As we will discuss later, this is the basis of JPEG.

Table 27-1b shows CS&Q to be a fixed-input fixed-output scheme. That is, a fixed number of bits are read from the input file and a smaller fixed number of bits are written to the output file. Other compression methods allow a variable number of bits to be read or written. As you go through the description of each of these compression methods, refer back to this table to understand how it fits into this classification scheme. Why are JPEG and MPEG not listed in this table? These are composite algorithms that combine many of the other techniques. They are too sophisticated to be classified into these simple categories.